Fiscal Year:	FY 2017	**Task Last Updated:**	FY 01/20/2017
PI Name:	Sams, Clarence Ph.D.		
Project Title:	Integration of Mechanotransduction and T-cell Activation Thresholds: Understanding of the Effects of Mechanical Forces on Assembly and Integration of Signal Transduction Machinery During T-cell Activation		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture		
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City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	10/01/2013	**End Date:**	10/01/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/01/2017 for reporting purposes; ok per F. Hernandez/ARC (Ed., 1/23/17) NOTE: Extended to 9/30/2017 per F. Hernandez/ARC (Ed., 9/29/16)		
Key Personnel Changes/Previous PI:	March 2016 report: Simon J. Clemett, Ph.D., senior planetary scientist, was added as CoInvestigator to do the digital image enhancement and multi-spectral rendering of the confocal images.		
COI Name (Institution):	Crucian, Brian Ph.D. (NASA Johnson Space Center) Nelman-Gonzalez, Mayra B.A. (Wyle Laboratories, Inc.) Clemett, Simon Ph.D. (ERC, Inc)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The influence of mechanical loading on cell structure and behavior is on the cutting edge of current cell biology research. Space flight has been postulated to alter mechanical loads on cells and it is known to alter cellular behavior. T-cell activation is known to be affected during space flight, but the association with altered mechanical environments remains unclear. Our hypothesis is that alterations in the mechanical environment to which T-cell are exposed can directly modulate the cell's ability to polarize and to integrate the signal transduction machinery. This results in a shift in the activation threshold due to altered assembly and effectiveness of cellular signaling systems. To characterize this phenomenon, primary T-cells will be activated in a cell-contact dependent manner under a variety of mechanical loading scenarios. Culture conditions include standard 1xG culture conditions, clinorotation (a ground-based culture model that provides elements of the physical environment present in cell culture during space flight), and controlled induction of mechanical loads. These experiments specifically will assess: 1) the effect of the mechanical forces on the activation response of T-cells, 2) morphological assessment of cell contact and polarization by environmental scanning electron microscopy/ESEM, 3) assessment of the integration of the mechanosensing and signaling components of the cell using confocal microscopy, and 4) proteomic assessment of early and downstream signal transduction events. This approach represents a novel application of biochemical, morphological, and mechanical assessment of T-cell activation and will provide a significant insight into the behavior of T-cells under changing environmental conditions. A more fundamental understanding of how T-cells integrate the signals during activation and how the external mechanical environment affects the threshold of activation will provide a mechanistic framework for interpretation of space flight and Earth-based immunology studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A more fundamental understanding of how T-cells integrate the signals during activation and how the external mechanical environment affects the threshold of activation will provide a mechanistic framework for interpretation of space flight and Earth-based immunology studies.
Task Progress:	Previous studies have demonstrated the importance of cell attachment, cell shape, and mechanical forces in modulating cell behavior. T cell activation experiments have shown a significant link exists between the mechanosensing of the T cell with its external environment and the integration of activating signals. To better understand the effects of altered mechanical environments on contact-mediated T cell activation, we used purified human peripheral T cells activated with CD3/CD28 immobilized on beads that induce attachment, cytoskeletal rearrangements, and signal transduction processes similar to those observed during stimulation by an antigen presenting cell (APC); modulation of the mechanical environment was achieved using a rotational suspension culture, which enables cell-bead contact while excluding contact with other surfaces. This culture-based system allowed us to examine the effects of altered mechanical forces on T cell polarization, cytoskeletal post-translational modifications (PTMs), spatial organization of key signaling proteins, and activation. Overall, cytoskeletal reorganization and polarization were significantly impaired during suspension culture. Examination of cell-bead constructs of T cells activated in suspension showed that although T cells contact and adhere to the beads, they remain round and do not exhibit extensive physical reorientation toward the contact site. Measurements of the cell-bead contact surface area showed a dramatic reduction of size in cells activated in suspension compared to static controls. Assessment of tubulin PTMs showed an increase in detyrosinated (Glu) microtubules in cells activated under suspension which suggests a reduction in cytoskeletal remodeling that can impact the transportation of proteins to their activating sites. T cells also required significantly more stimulation to activate in suspension culture as compared to static culture (increased activation threshold). This change in threshold involves a mechanism independent of TCR (T-cell receptor) triggering as T cells stimulated during suspension required 2- to 3-fold higher levels of TCR triggering to achieve equivalent levels of activation compared to static cells. Evaluation of the integration of the signal transduction pathways showed that ZAP 70 and Erk1/Erk2 phosphorylation was significantly inhibited in T cells activated in suspension culture. In addition, spatial reorganization and clustering of both Zap 70 and Erk1/Erk2 did not occur in the suspension activated T cells as compared to static controls. Our results suggest that mechanical forces play a primary role in the regulation and integration of signal transduction networks necessary for successful T cell activation.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Abstracts for Journals and Proceedings	Nelman-Gonzalez MA, Clemett SJ, Beitman AJ, Alexander MR, Merrill CL, Crucian BE, Sams CF. "Integration of mechanotransduction and T-cell activation thresholds: the effects of mechanical forces on assembly and integration of the signal transduction machinery during T cell activation." Presented at the 2016 Annual Meeting of the American Society for Cell Biology, San Francisco, CA, December 3-7, 2016. Molecular Biology of the Cell. 2016 Dec 15;27(25):3947. http://dx.doi.org/10.1091/mbc.E16-10-0736 , Dec-2016